Equation of motion

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\} + \delta x_D[k],$$

Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{f_{dx}[k] + f_T[k]\} - \delta x_D[k],$$

$\delta x[k] = x_d[k] - x[k]$, $x_d[k]$ is the desired motion trajectory, $\delta x_D[k]$ is the motion error caused by the disturbance, and $\Delta x_d[k] = x_d[k+1] - x_d[k]$ is the desired motion step size.

Measurement

(d-step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k],$$

Control objective: $\delta x[k+1] = \lambda_c \cdot \delta x[k]; \quad 0 \leq \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dx}[k] = a_x^{-1}[k]\{\Delta x_d[k] + (1-\lambda_c)\,\boldsymbol{\delta x[k]} - \delta x_D[k]\}$$

Motion error:

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{f}_{dx}[k] + f_T[k]\} - \delta x_D[k] = \lambda_c \cdot \delta x[k] - a_x[k] f_T[k]$$

d-step measurement delay:

$$\boldsymbol{\delta x[k]} = \delta x[k-d] + \sum_{i=1}^{d} \Delta x_d[k-i] - \sum_{i=1}^{d} a_x[k-i]\{f_{dx}[k-i] + f_T[k-i]\} - \sum_{i=1}^{d} \delta x_D[k-i]$$

Ideal control effort (with d-step measurement delay):

$$\bar{\bar{f}}_{dx}[k] = a_x^{-1}[k]\Bigg\{ \underbrace{\Big[\Delta x_d[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} \Delta x_d[k-i]\Big]}_{\Delta x_d^d[k]} + (1-\lambda_c)\Big[\delta x[k-d] - \textstyle\sum_{i=1}^{d} a_x[k-i] f_{dx}[k-i]\Big]$$
$$- \underbrace{\Big[\delta x_D[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} \delta x_D[k-i]\Big]}_{\delta x_D^d[k]} \Bigg\}$$

Motion error:

$$\begin{aligned}\delta x[k+1] &= \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{\bar{f}}_{dx}[k] + f_T[k]\} - \delta x_D[k] \\ &= \lambda_c \cdot \delta x[k] - \underbrace{\Big\{a_x[k] f_T[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} a_x[k-i] f_T[k-i]\Big\}}_{\delta x_T^d[k]}\end{aligned}$$

Implementable control effort

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k]\Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x_m[k] - \sum_{i=1}^{d} \hat{a}_x[k-i] f_{dx}[k-i]\Big] - \delta\hat{x}_D^d[k]\Big\}}$$

Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k]\big[f_{dx}[k] - \bar{\bar{f}}_{dx}[k]\big] - \delta x_T^d[k]$$

$$a_x[k]\,\bar{\bar{f}}_{dx}[k] = \Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x[k-d] - \sum_{i=1}^{d} a_x[k-i] f_{dx}[k-i]\Big] - \delta x_D^d[k]\Big\}$$

$$\begin{aligned} a_x[k]\,f_{dx}[k] = \big(1 + e_{a_x}[k]\,\hat{a}_x^{-1}[k]\big)\Big\{&\Delta x_d^d[k] \\ &+ (1-\lambda_c)\Big[\big(\delta x[k-d] + n_x[k]\big) - \sum_{i=1}^{d} \hat{a}_x[k-i] f_{dx}[k-i]\Big] - \delta\hat{x}_D^d[k]\Big\} \end{aligned}$$

$$\begin{aligned} a_x[k]\,f_{dx}[k] - a_x[k]\,\bar{\bar{f}}_{dx}[k] = e_{a_x}[k]\,f_{dx}[k] + (1-\lambda_c) n_x[k] \\ + (1-\lambda_c)\sum_{i=1}^{d}\big(a_x[k-i] - \hat{a}_x[k-i]\big) f_{dx}[k-i] + e_{\delta x_D^d}[k] \end{aligned}$$

Parameter model:

$$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]; \quad a_{x_{ram}}[k] \approx a'_x[k]\,x_{ram}[k]$$

$$\begin{aligned} a_x[k+1] &= \boldsymbol{a_{x_{det}}[k+1]} + a_{x_{ram}}[k+1] \\ &= \underbrace{\Big\{\boldsymbol{a_{x_{det}}[k]} + a_{x_{ram}}[k]\Big\}}_{a_x[k]} + \boldsymbol{\delta a[k]} + \underbrace{\Big\{a_{x_{ram}}[k+1] - a_{x_{ram}}[k]\Big\}}_{\delta a_{ram}[k]} \end{aligned}$$

$$\begin{cases} a_x[k+1] = a_x[k] + \delta a_x[k] + \delta a_{ram}[k] \\ \delta a_x[k+1] = \delta a_x[k] \end{cases}$$

Parameter estimator:

$$\begin{cases} \hat{a}_x[k+1] = \hat{a}_x[k] + \delta\hat{a}_x[k] \\ \delta\hat{a}_x[k+1] = \delta\hat{a}_x[k] \end{cases}$$

Parameter estimation error:

$$\begin{cases} e_{a_x}[k] = a_x[k] - \hat{a}_x[k] \\ e_{\delta a_x}[k] = \delta a_x[k] - \delta\hat{a}_x[k] \end{cases}$$

Estimation error dynamics:

$$\begin{cases} e_{a_x}[k+1] = e_{a_x}[k] + e_{\delta a_x}[k] + \delta a_{ram}[k] \\ e_{\delta a_x}[k+1] = e_{\delta a_x}[k] \end{cases}$$

$$a_x[k-i] - \hat{a}_x[k-i] \approx e_{a_x}[k] - i \cdot e_{\delta a_x}[k] - \sum_{j=1}^{i} \delta a_{ram}[k-j]$$

$$\begin{aligned}
a_x[k]\, f_{dx}[k] - a_x[k]\, \bar{\bar{f}}_{dx}[k] \approx e_{a_x}[k]\ &\underbrace{\left\{ f_{dx}[k] + (1-\lambda_c) \textstyle\sum_{i=1}^{d} f_{dx}[k-i] \right\}}_{F_{dx}[k]} \\
&- e_{\delta a_x}[k]\ \underbrace{\left\{ (1-\lambda_c) \textstyle\sum_{i=1}^{d} i \cdot f_{dx}[k-i] \right\}}_{dF_{dx}[k]} \\
&- \underbrace{\left\{ (1-\lambda_c) \textstyle\sum_{i=1}^{d} (d+1-i) \cdot f_{dx}[k-i]\, \delta a_{ram}[k-i] \right\}}_{\delta x^d_{\delta af}[k]} \\
&+ e_{\delta x^d_D}[k] + (1-\lambda_c) n_x[k]
\end{aligned}$$

$$\boxed{\begin{aligned}
\delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] + dF_{dx}[k] \cdot e_{\delta a_x}[k] - e_{\delta x^d_D}[k] \\
- \underbrace{\left\{ \delta x^d_T[k] - \delta x^d_{\delta af}[k] + (1-\lambda_c) n_x[k] \right\}}_{\varepsilon[k]}
\end{aligned}}$$

State equation

$$\begin{aligned}
\delta x_1[k+1] &= \delta x_2[k] \\
\delta x_2[k+1] &= \delta x_3[k] \\
\delta x_3[k+1] &= \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] + dF_{dx}[k] \cdot e_{\delta a_x}[k] - e_{\delta x^d_D}[k] - \varepsilon[k] \\
\delta x^d_D[k+1] &= \delta x^d_D[k] \\
a_x[k+1] &= a_x[k] + \delta a_x[k] + \delta a_{ram}[k] \\
\delta a_x[k+1] &= \delta a_x[k]
\end{aligned}$$

Output equation

$$\begin{cases}
y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\
y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} \approx a_x[k] - d \cdot \delta a_x[k] + \underbrace{\left\{ n_a[k] - \textstyle\sum_{i=1}^{d} \delta a_{ram}[k-i] \right\}}_{r_2[k]}
\end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -d \end{bmatrix}$$

$$\mathbf{y}[k] = \mathbf{H}\mathbf{x}[k] + \mathbf{r}[k]$$

Closed-loop Estimator

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta\hat{x}_1[k]$$

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{\delta a_x}[k] = \delta a_x[k] - \delta\hat{a}_x[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = \delta x_m[k] - \delta\hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$\begin{aligned} e_{y_2}[k] &= y_2[k] - \hat{y}_2[k] = a_{xm}[k] - \hat{a}_x[k] + d \cdot \delta\hat{a}_x[k] \\ &= e_{a_x}[k] - d \cdot e_{\delta a_x}[k] + \underbrace{\left\{ n_a[k] - \textstyle\sum_{i=1}^{d} \delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{aligned}$$

$$\begin{aligned} \delta\hat{x}_1[k+1] &= \delta\hat{x}_2[k] + \ell_{11}[k]\, e_{y_1}[k] + \ell_{12}[k]\, e_{y_2}[k] \\ \delta\hat{x}_2[k+1] &= \delta\hat{x}_3[k] + \ell_{21}[k]\, e_{y_1}[k] + \ell_{22}[k]\, e_{y_2}[k] \\ \delta\hat{x}_3[k+1] &= \lambda_c \cdot \delta\hat{x}_3[k] + \ell_{31}[k]\, e_{y_1}[k] + \ell_{32}[k]\, e_{y_2}[k] \\ \delta\hat{x}_D^d[k+1] &= \delta\hat{x}_D^d[k] + \ell_{41}[k]\, e_{y_1}[k] + \ell_{42}[k]\, e_{y_2}[k] \\ \hat{a}_x[k+1] &= \hat{a}_x[k] + \delta\hat{a}_x[k] + \ell_{51}[k]\, e_{y_1}[k] + \ell_{52}[k]\, e_{y_2}[k] \\ \delta\hat{a}_x[k+1] &= \delta\hat{a}_x[k] + \ell_{61}[k]\, e_{y_1}[k] + \ell_{62}[k]\, e_{y_2}[k] \end{aligned}$$

Error dynamics

$$\begin{aligned} e_{\delta x_1}[k+1] &= e_{\delta x_2}[k] - \ell_{11}[k]\, e_{\delta x_1}[k] - \ell_{12}[k]\, e_{a_x}[k] + \ell_{12}[k] \cdot d \cdot e_{\delta a_x}[k] \\ e_{\delta x_2}[k+1] &= e_{\delta x_3}[k] - \ell_{21}[k]\, e_{\delta x_1}[k] - \ell_{22}[k]\, e_{a_x}[k] + \ell_{22}[k] \cdot d \cdot e_{\delta a_x}[k] \\ e_{\delta x_3}[k+1] &= \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k] \cdot e_{a_x}[k] + dF_{dx}[k] \cdot e_{\delta a_x}[k] - e_{\delta x_D^d}[k] \\ &\qquad - \ell_{31}[k]\, e_{\delta x_1}[k] - \ell_{32}[k]\, e_{a_x}[k] + \ell_{32}[k] \cdot d \cdot e_{\delta a_x}[k] - \underbrace{\varepsilon[k]}_{q_3[k]} \\ e_{\delta x_D^d}[k+1] &= e_{\delta x_D^d}[k] - \ell_{41}[k]\, e_{\delta x_1}[k] - \ell_{42}[k]\, e_{a_x}[k] + \ell_{42}[k] \cdot d \cdot e_{\delta a_x}[k] \\ e_{a_x}[k+1] &= e_{a_x}[k] + e_{\delta a_x}[k] - \ell_{51}[k]\, e_{\delta x_1}[k] - \ell_{52}[k]\, e_{a_x}[k] + \ell_{52}[k] \cdot d \cdot e_{\delta a_x}[k] + \underbrace{\delta a_{ram}[k]}_{q_5[k]} \\ e_{\delta a_x}[k+1] &= e_{\delta a_x}[k] - \ell_{61}[k]\, e_{\delta x_1}[k] - \ell_{62}[k]\, e_{a_x}[k] + \ell_{62}[k] \cdot d \cdot e_{\delta a_x}[k] \end{aligned}$$

$$\boxed{\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -1 & -F_{dx}[k] & dF_{dx}[k] \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}}$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}\,\mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}\,\mathbf{r}[k]$$